

Mixed Review: Logarithms → Polynomials

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***Standard:** CCSS.MATH.CONTENT.HSF-BF.B.5 / HSN-CN.C.9 — Apply the inverse relationship between exponents and logarithms; extend polynomial identities to the complex numbers using the Fundamental Theorem of Algebra.*

Part 1 — Logarithms (4 problems)

Directions: Show every step. For Problem 4, state the domain restriction and reject any extraneous solutions.

1. Use the change-of-base formula to evaluate $\log_5(28)$. Give the *exact* form (as a quotient of common or natural logs) and a decimal approximation rounded to 4 decimal places.

2. Expand fully using the laws of logarithms (no logs of products, quotients, or powers should remain):

$$\log_3 \left(\frac{x^2 \sqrt{y}}{z^4} \right)$$

3. Condense into a single logarithm:

$$3 \ln(x) - \frac{1}{2} \ln(y+1) + 2 \ln(z)$$

4. Solve for x . State the domain first, solve algebraically, and reject any extraneous solutions.

$$\log_2(x-3) + \log_2(x+1) = 5$$

Part 2 — Exponential Modeling (3 problems)

Directions: Set up the model first, then solve. Round final numerical answers to 2 decimal places unless otherwise stated.

5. A principal of \$4,500 is invested at an annual rate of 4.2%, **compounded continuously**. Using $A(t) = Pe^{rt}$, determine how many *years* it will take for the investment to grow to \$10,000.

6. A radioactive sample decays according to $A(t) = A_0 e^{kt}$, where t is in years.
(a) If the sample's **half-life is 28 years**, find the exact value of k (leave it in terms of $\ln$).
(b) Using your value of k , find how many years it takes until only 12% of the original sample remains.

7. Convert between exponential bases. Rewrite $f(t) = 600 (1.035)^t$ in the form $f(t) = 600 e^{rt}$.
(a) Find the exact value of r .
(b) Interpret r as a continuous growth rate (state as a percentage to 2 decimal places).

Part 3 — Fundamental Theorem of Algebra (2 problems)

Directions: Use the Conjugate Root Theorem. Show your division work.

8. Given that $(2 - i)$ is a zero of $P(x) = x^4 - 5x^3 + 7x^2 + 3x - 10$, find **all four zeros** and write $P(x)$ in fully factored linear form.

9. Find a polynomial $P(x)$ of degree 4 with **real integer coefficients** having zeros $x = 3$ (multiplicity 2) and $x = 1 + 2i$. Expand your answer to standard form $ax^4 + bx^3 + cx^2 + dx + e$.

Part 4 — Synthetic Division & Factor Theorem (2 problems)

Directions: Use synthetic division. For #10, fill in missing terms with zero coefficients.

10. Use synthetic division to divide $(3x^4 - 8x^3 + 5x - 7) \div (x - 2)$. State the quotient $Q(x)$ and remainder R . Then verify your remainder by computing $P(2)$ directly (Remainder Theorem).

11. Find the value of k so that $(x + 4)$ is a factor of $P(x) = 2x^3 + kx^2 - 11x - 12$. Justify your answer using the Factor Theorem.

Part 5 — Polynomial Graphing (1 problem)

Directions: Use multiplicity rules (bounce / cross / slide) and end-behavior analysis.

- 12.** Given $P(x) = -(x + 1)^2(x - 2)^3(x - 4)$:
- (a) State the **degree** and the **end behavior** (use arrow notation $\uparrow / \downarrow$).
 - (b) List all **zeros** with their **multiplicities** and classify each as *bounce*, *cross*, or *slide*.
 - (c) Find the y -intercept.
 - (d) Sketch a rough graph showing all the above features clearly.

